

Study Report: Continuous Delivery - QA & Test Engineer

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Family: QA & Test Engineer

Level: Advanced

Chapters: 7

Source: Reference: Continuous Delivery - Jez Humble & David Farley

Official sources and free libraries

- <https://continuousdelivery.com/>

Summary

Study report for **Continuous Delivery** by Jez Humble & David Farley (Advanced level) mapped to the QA & Test Engineer role. CI/CD pipelines, test automation gates, and deployment safety.

Chapters

Track: Book-Intro

Chapter 1: About Continuous Delivery [Advanced]

Chapter focus: About Continuous Delivery** ***(Level: Advanced)**

This study report summarizes **Continuous Delivery** by Jez Humble & David Farley for the QA & Test Engineer role. The resource is rated Advanced level. CI/CD pipelines, test automation gates, and deployment safety. Use this report alongside the official material to prepare for CodeQuest review.

Code Reference:

```
# Continuous Delivery
# Author: Jez Humble & David Farley
# Role: QA & Test Engineer
# Level: Advanced
```

What it shows: Metadata block records the source book and target role family.

Topic guide

What it actually is

A structured study report based on Continuous Delivery.

When to use it

When learning QA & Test Engineer skills at Advanced level.

Where to use it

QA & Test Engineer training paths and certification prep.

Real use example

A learner reads this report before starting the full book or free online guide.

Track: Book-Context

Chapter 2: Why This Book Matters for Your Role [Advanced]

Chapter focus: Why This Book Matters for Your Role *(Level: Advanced)**

QA & Test Engineer professionals use ideas from Continuous Delivery to solve real workplace problems. CI/CD pipelines, test automation gates, and deployment safety. This chapter explains how the book fits into your learning path and what you should be able to do after studying it.

Code Reference:

Role: QA & Test Engineer

Book focus: CI/CD pipelines, test automation gates, and deployment safety.

Recommended level: Advanced

What it shows: Connects the book topic to the job role outcomes.

Topic guide

What it actually is

Role-aligned learning connects theory to job tasks.

When to use it

During career planning and syllabus design.

Where to use it

QA & Test Engineer bootcamps and CodeQuest teacher assignments.

Real use example

A teacher assigns this book report before a advanced skills checkpoint.

Track: Book-Topics

Chapter 3: Key Topics Covered [Advanced]

Chapter focus: Key Topics Covered *(Level: Advanced)**

The main topics in Continuous Delivery include practical concepts described as: CI/CD pipelines, test automation gates, and deployment safety. Study each topic with hands-on practice. Take notes on definitions, workflows, and examples that match tools used in QA & Test Engineer jobs today.

Code Reference:

- Topic 1: core concept
- Topic 2: core concept
- Topic 3: core concept
- Topic 4: core concept

What it shows: Topic list guides what to study chapter-by-chapter in the source.

Topic guide**What it actually is**

Topic maps turn a book into actionable learning objectives.

When to use it

Before reading and while building chapter summaries.

Where to use it

Self-study, flipped classroom, and revision.

Real use example

A student maps each book chapter to a CodeQuest report section.

Track: Book-Applied

Chapter 4: Applied: Performance Testing with k6 [Advanced]**Chapter focus: Performance Testing with k6** *(Level: Advanced)**

Performance testing validates latency and throughput under expected and peak load. k6 scripts define VUs (virtual users) ramping scenarios in JavaScript. Thresholds fail CI when P95 exceeds SLA (e.g., `http_req_duration p(95)<500ms`). Test staging only with approval-never DDoS production inadvertently.

Code Reference:

```
import http from 'k6/http';
import { check } from 'k6';
export default function () {
  const res = http.get('https://staging.example/api/products');
  check(res, { 'status 200': (r) => r.status === 200 });
}
```

What it shows: Virtual user iteration hits products endpoint; check validates status.

Topic guide

What it actually is

Load testing reveals scalability bottlenecks before users do during peaks.

When to use it

Run before major sales events and after infrastructure changes.

Where to use it

E-commerce flash sales, election result pages, and API gateway sizing.

Real use example

k6 finds DB connection pool exhausted at 200 VUs; pool size increased pre-launch.

Chapter 5: Applied: ISTQB Test Planning and Metrics [Advanced]

Chapter focus: ISTQB Test Planning and Metrics** *(Level: Advanced)

Test plan documents scope, approach, resources, schedule, and exit criteria. Entry criteria define when testing starts; exit criteria define release readiness. Defect density (bugs/KLOC) and escape rate (prod bugs/release) track quality trends. Risk-based testing prioritizes effort on high-impact, high-likelihood areas.

Code Reference:

Exit Criteria Example:

- 100% critical/high test cases executed
- Zero open P1 defects
- <=2 open P2 with business sign-off
- Automation smoke pass on release candidate build

What it shows: Exit criteria make go/no-go objective rather than subjective gut feel.

Topic guide

What it actually is

Formal test planning aligns QA activity with business risk tolerance.

When to use it

Required in regulated domains; valuable everywhere releases are frequent.

Where to use it

Medical devices, fintech, and enterprise RFP-driven projects.

Real use example

Release board approves ship when exit criteria met; one waived P2 documented.

Chapter 6: Applied: Flaky Test Triage and Quarantine [Advanced]

Chapter focus: Flaky Test Triage and Quarantine** *(Level: Advanced)

Flaky tests pass and fail without code changes-eroding trust in CI. Quarantine flaky tests (@pytest.mark.quarantine) off critical path while fixing root cause. Common causes: timing, shared state, external dependency instability, clock sensitivity. Track flake rate metric; zero tolerance on main branch smoke suite.

Code Reference:

```
# Retry only in CI with pytest-rerunfailures
# @pytest.mark.flaky(reruns=2) - temporary, must ticket root cause
```

What it shows: Reruns mitigate temporarily but root cause fix required for permanent trust.

Topic guide

What it actually is

Flake management keeps CI signal trustworthy for release decisions.

When to use it

Implement quarantine policy when E2E suite exceeds 50 tests.

Where to use it

Large Playwright/Selenium suites in high-velocity teams.

Real use example

Race on toast notification fixed with expect().to_be_visible() replacing sleep.

Track: Book-Plan

Chapter 7: Study Plan and Practice [Advanced]

Chapter focus: Study Plan and Practice** *(Level: Advanced)

Finish Continuous Delivery with a weekly plan: read one section, write a summary, complete one exercise, and reflect on how it applies to QA & Test Engineer. If a free edition exists, practice every example. Submit your notes and one mini-project demo for teacher review.

Code Reference:

```
Week 1: Read + notes
Week 2: Exercises
Week 3: Mini project
Week 4: Review + quiz
```

What it shows: A 4-week plan turns reading into demonstrable skill.

Topic guide

What it actually is

Structured study plans improve retention and portfolio outcomes.

When to use it

After finishing the key topics chapters.

Where to use it

CodeQuest end-of-unit assessments.

Real use example

A learner completes the study plan and uploads a advanced project screenshot.